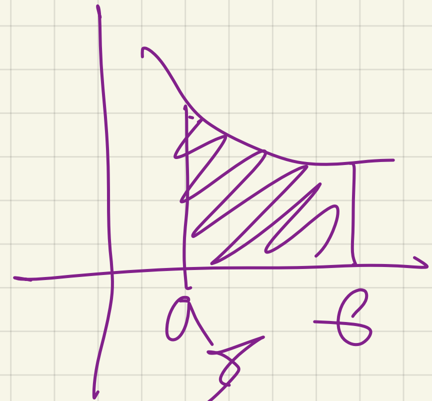
$y = f(x)$

$\underbrace{F(b) - F(a)} = \underbrace{\int_a^b f(x)\,dx}$



$\int_b^{\delta} f(x)\,dx$

$a = 1$ $\quad h = 8 \quad S = \left(\frac{1+2}{2}\right)\cdot 8 = 12$

$b = 2$

---

1 2008 1000р 20%

2014 2200 44%

C:

(1) $\underbrace{1000 + 1000\cdot 1{,}2 + 1000\cdot 1{,}2^2 + \ldots + 1000\cdot 1{,}2^n}_{n+1}$

(2) $2200 + \ldots + 2200\cdot 1{,}44^k$

$k = n - 6$

$$\sum_1 = 1000(1 + 1{,}2 + \ldots + 1{,}2^n) =$$

$$1000 \cdot \left(\frac{1{,}2^{n+1} - 1}{1{,}2 - 1} \right) = \frac{1000}{0{,}2}(1{,}2^{n+1} - 1)$$

$$5000(1{,}2^{n+1} - 1)$$

$$\sum_2 = 2200\left(1 + \ldots + 1{,}44^{n-6}\right) =$$

$$= 2200\left(\frac{1{,}44^{n-5} - 1}{1{,}44 - 1}\right) = \frac{2200}{0{,}44}(1{,}44^{n-5} - 1)$$

$$= 5000(1{,}44^{n-5} - 1)$$

$$5000(1{,}2^{n+1} - 1) = 5000(1{,}44^{n-5} - 1)$$

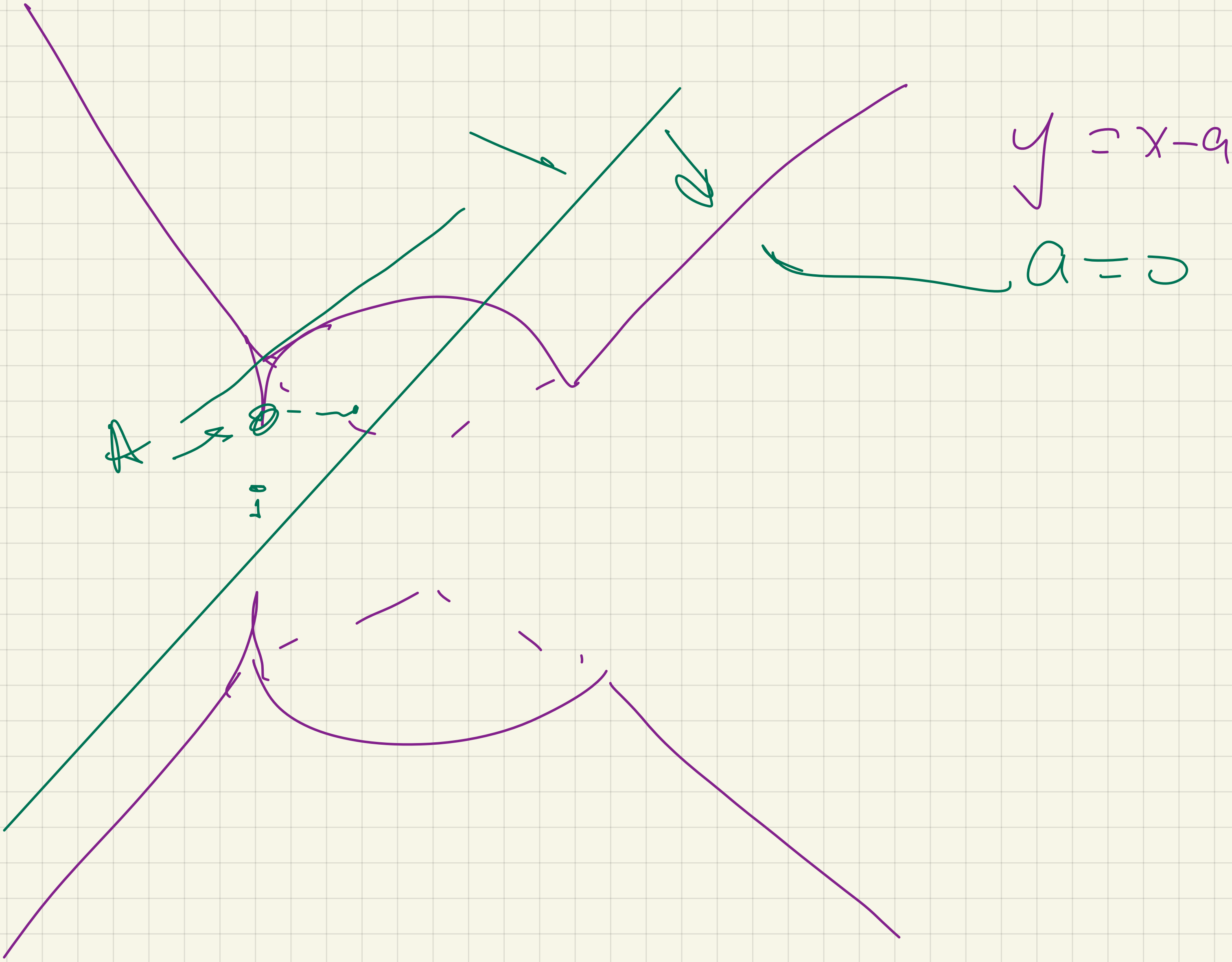
$$n = ?$$

$$1{,}2^{n+1} = 1{,}44^{n-5}$$

$$1{,}2^{n+1} = (1{,}2)^{2(n-5)}$$

$$n+1 = 2n - 10$$

$$n = 11$$



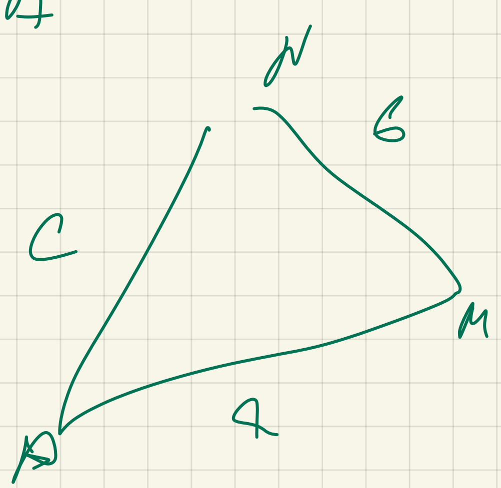
$BM = \sqrt{41}$ +

MM и SA

MM MN

MN

BN →



$$a^2+b^2-2ab\cos\alpha=c^2$$

$$\begin{cases} \cos(y-a)-2\cos x=0 \\ \log_2(ay-y^2)=2\log_4(-x)-\log_{1/2}(3y) \end{cases}$$

$\underline{\underline{2n+1}}$ реш $n\in\mathbb{Z}$

$$\log_2(ay-y^2)=\cancel{\frac{2}{2}}\log_2(-x)+\log_2(3y)$$

$$\cancel{\log_2 y}+\log_2(a-y)=\log_2(-x)+\log_2 3+\cancel{\log_2 y}$$

$$\begin{cases} \log_2(a-y)=\log_2(-3x) \\ y>0 \end{cases}$$

$a-y>0$

$$\begin{cases} \underline{a-y}=-\underline{\underline{3x}}>0 \quad y<a \\ x<0 \\ y>0 \end{cases}$$

$$\cos(y-a) = 2\cos x$$

$$\cos(-3x) = 2\cos x$$

$$\cos 3x = 2\cos x$$

$$\cos 3x = 4\cos^3 x - 3\cos x$$

$$4\cos^3 x - 3\cos x = 2\cos x$$

$$4\cos^3 x - 5\cos x = 0$$

$$\cos x(4\cos^2 x - 5) = 0$$

$$\left[\begin{array}{l} \cos x = 0 \longrightarrow \\ \cos^2 x = \frac{5}{4} \quad \varnothing \end{array}\right.$$

$$\cos x = 0 \quad \rightarrow \quad x = \frac{\pi}{2} + \pi k$$

$$y = a + 3x$$

$$a = 0$$

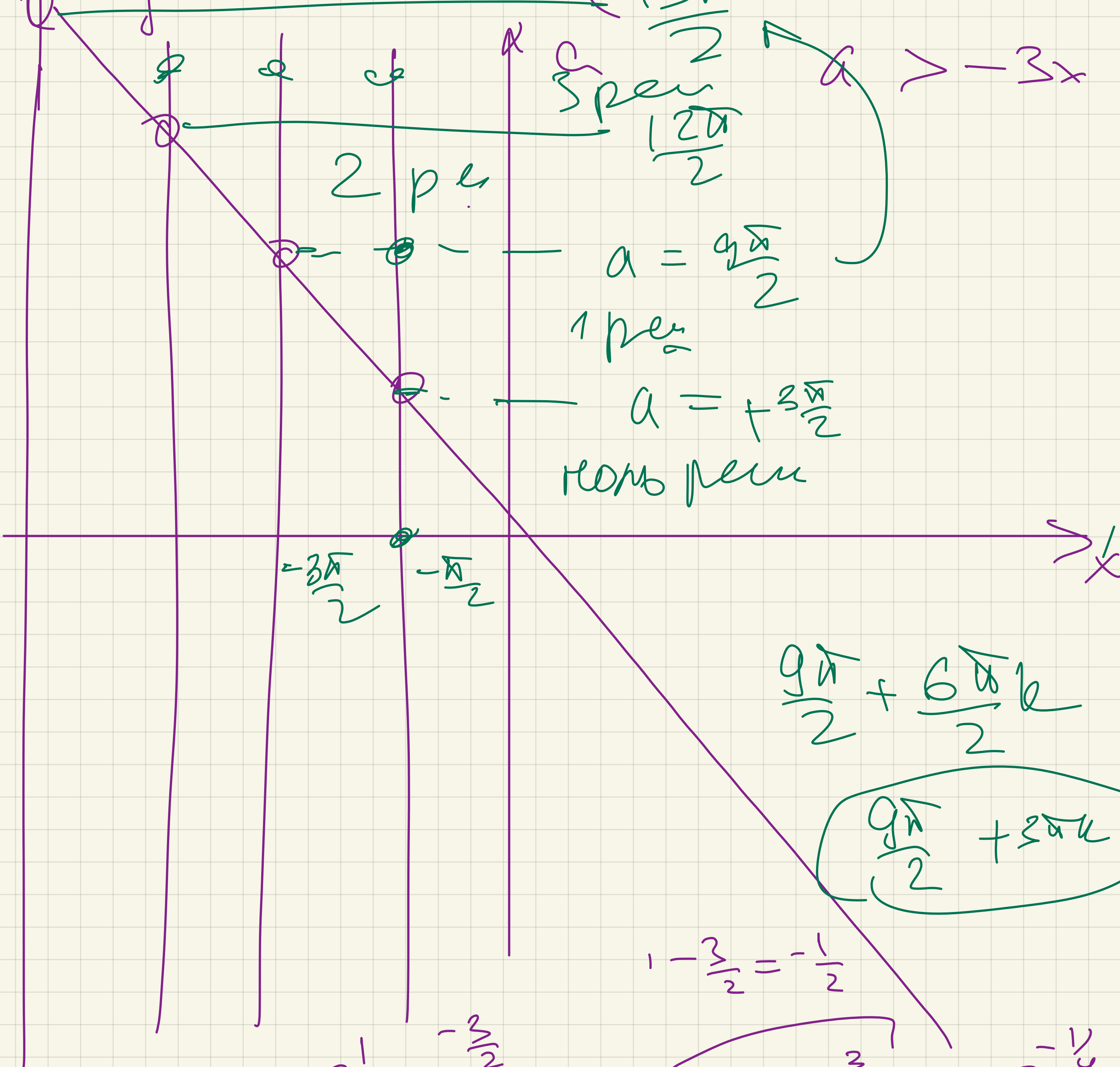
$k = 0$

$x = \frac{\pi}{2}$

$k = -1$

$-\frac{\pi}{2}$

$$\left[\begin{cases} x = \frac{\pi}{2} + \pi k \\ a - y = -3x \\ x < 0 \\ y > 0 \end{cases}\right. \qquad \begin{cases} x = \frac{\pi}{2} - \pi k \\ y = 3x + a > 0 \\ y > 0 \end{cases}$$



$\frac{9\pi}{2} + \frac{6\pi k}{2}$

$\frac{9\pi}{2} + 3\pi k$

$1 - \frac{3}{2} = -\frac{1}{2}$

$3^{1} \; 3^{-\frac{3}{2}}$

$\sqrt{3|3|^{-\frac{3}{2}}} = 3^{-\frac{1}{4}}$

$$\begin{cases} y^2 - x^2 \geq 0 \\ (y - a^2 - 3a + 18)^2 + (x - 6a)^2 = 3|a|^{-\frac{a}{2}} \end{cases}$$

$(6a;\ a^2+3a-18)$

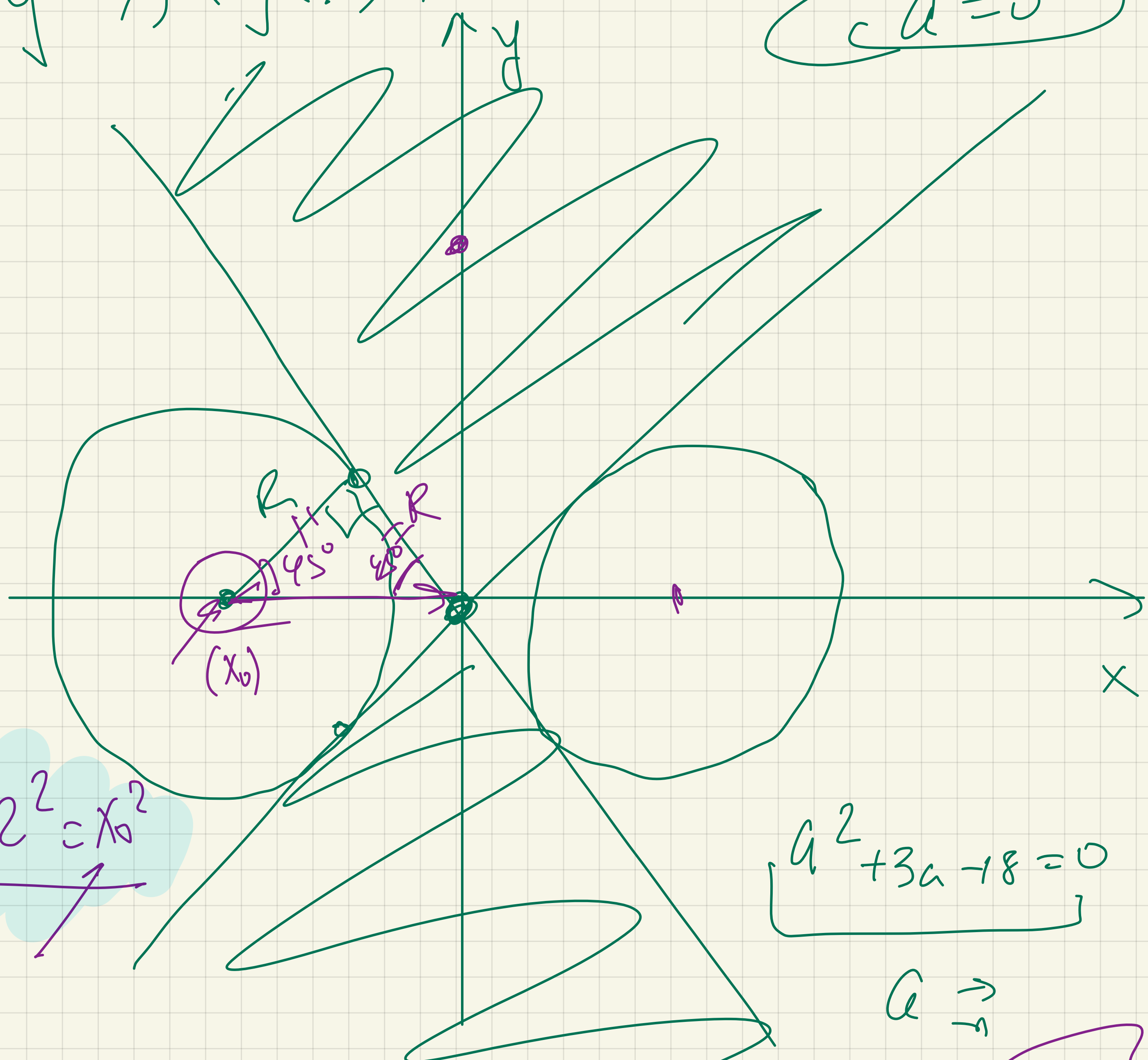
$R = \sqrt{3|a|^{-\frac{a}{2}}}$

2 реш

$(0;\ 18)$

$(y-x)(y+x) \geq 0$

$a = 0$



$R^2 + R^2 = x_0^2$

$a^2 + 3a - 18 = 0$

$a \to$

$a = 3$

$R = 9\sqrt{2} \leftarrow$ gomnuen $\sqrt{3 \cdot 3^{-\frac{3}{2}}} = 3^{-\frac{1}{4}}$

$a = -6$ $R = 18\sqrt{2}$ $\rightarrow$

$3 \cdot |-6|^{\frac{3}{2}} =$

$\sqrt{3 \cdot 6^3} =$

$\sqrt{3^4 \cdot 2^3} =$

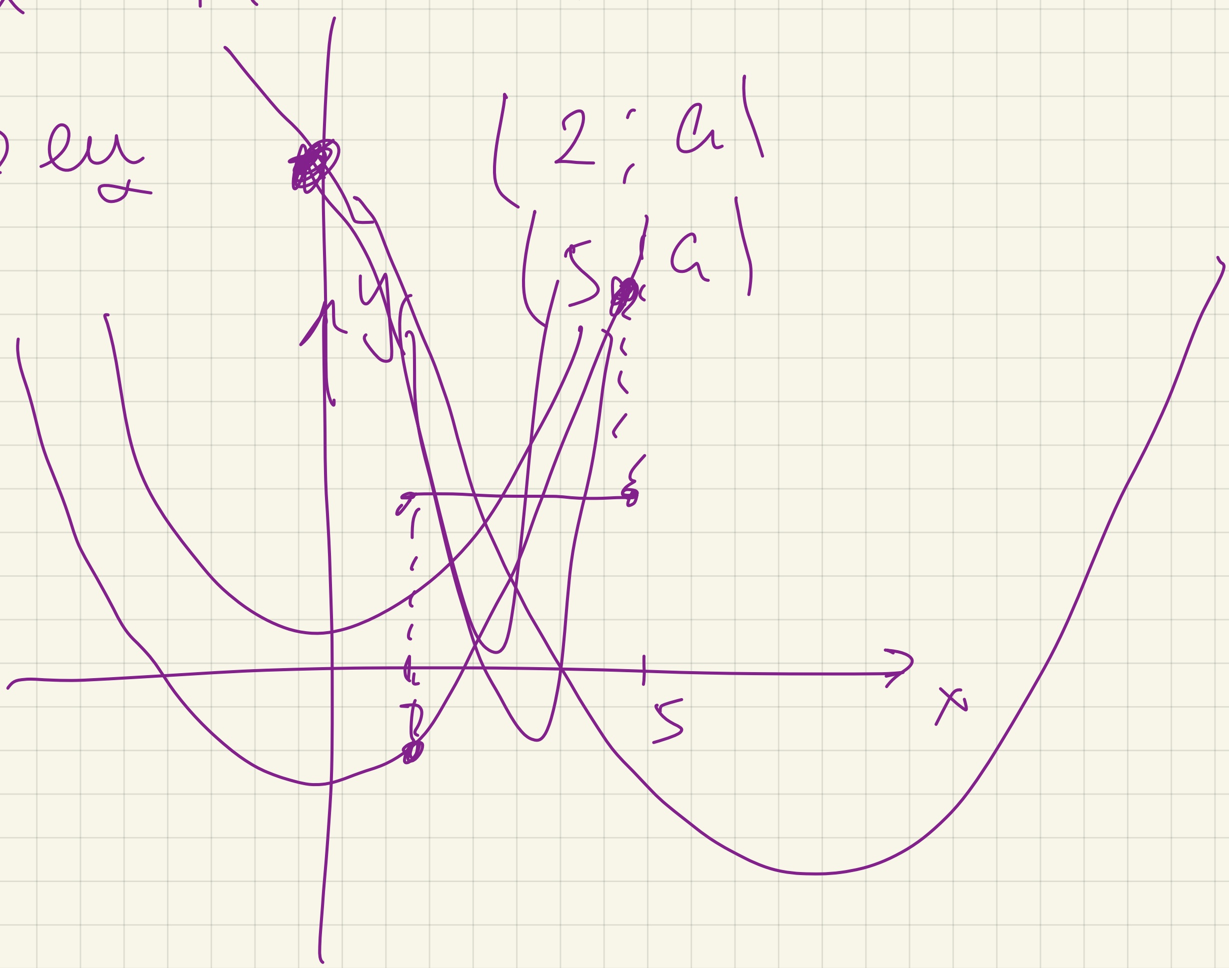
$3^2 \cdot 2\sqrt{2} = 18\sqrt{2}$

$$\begin{cases} \sqrt{(x-2)^2 + (y-a)^2} + \sqrt{(x-5)^2 + (y-a)^2} = 3 \\ x^2 - |a+2|x - 3a^2 = 5 \end{cases}$$

1 реш

$(2; a)$

$(5; a)$



$$f(x) = x^2 - (a+2)x - 3a^2 - 5$$

$$f(2) \leq 0$$

$$f(5) \geq 0$$

$$\begin{cases} 4 - (a+2)2 - 3a^2 - 5 \leq 0 \\ 25 - (a+2)5 - 3a^2 - 5 \geq 0 \end{cases}$$

$a \rightarrow$

$D/3$